

ECEN 662: Estimation & Detection Theory

Lecture 4: Uniformly Minimum-Variance Unbiased Estimation (Part I)

Tie Liu

Texas A&M University

Parameter estimation: The frequentist view

- Let θ be an *unknown* parameter, and let $g(\theta)$ be a function of θ . The goal of *parameter estimation* is to guess the value of $g(\theta)$ based on an observation of X , which is generated according to $f_\theta(x)$

Parameter estimation: The frequentist view

- Let θ be an *unknown* parameter, and let $g(\theta)$ be a function of θ . The goal of *parameter estimation* is to guess the value of $g(\theta)$ based on an observation of X , which is generated according to $f_{\theta}(x)$
- From the modeling viewpoint, there are two views on the unknown parameter θ : a *frequentist* view and a *Bayesian* view

Parameter estimation: The frequentist view

- Let θ be an *unknown* parameter, and let $g(\theta)$ be a function of θ . The goal of *parameter estimation* is to guess the value of $g(\theta)$ based on an observation of X , which is generated according to $f_\theta(x)$
- From the modeling viewpoint, there are two views on the unknown parameter θ : a *frequentist* view and a *Bayesian* view
- For our course, we shall focus on the *frequentist* view, in which θ is *deterministic* (but unknown)

Parameter estimation: The frequentist view

- Let θ be an *unknown* parameter, and let $g(\theta)$ be a function of θ . The goal of *parameter estimation* is to guess the value of $g(\theta)$ based on an observation of X , which is generated according to $f_\theta(x)$
- From the modeling viewpoint, there are two views on the unknown parameter θ : a *frequentist* view and a *Bayesian* view
- For our course, we shall focus on the *frequentist* view, in which θ is *deterministic* (but unknown)
- The observation X , on the other hand, is *random* and can be sampled *repeatedly*, i.e., $X = (X_1, \dots, X_n)$ where $X_1, \dots, X_n$ are independent measurements given θ

Parameter estimation: The frequentist view

- Let θ be an *unknown* parameter, and let $g(\theta)$ be a function of θ . The goal of *parameter estimation* is to guess the value of $g(\theta)$ based on an observation of X , which is generated according to $f_\theta(x)$
- From the modeling viewpoint, there are two views on the unknown parameter θ : a *frequentist* view and a *Bayesian* view
- For our course, we shall focus on the *frequentist* view, in which θ is *deterministic* (but unknown)
- The observation X , on the other hand, is *random* and can be sampled *repeatedly*, i.e., $X = (X_1, \dots, X_n)$ where $X_1, \dots, X_n$ are independent measurements given θ
- Because in the eye of a frequentist, the parameter θ is deterministic but unknown, the *performance* of an estimator $\hat{g} = \hat{g}(X)$ has to be assessed across *all* possible values of θ

- In addition, even though θ is deterministic, $\hat{g} = \hat{g}(X)$ is in general *random* because the observation X is random, so its performance has to be assessed in *probabilistic* terms for each given θ

Uniformly minimum-variance unbiased estimator

- In statistics, a classic notion of an *optimal* frequentist estimator is *uniformly minimum-variance unbiased estimator (UMVUE)*

Uniformly minimum-variance unbiased estimator

- In statistics, a classic notion of an *optimal* frequentist estimator is *uniformly minimum-variance unbiased estimator (UMVUE)*
- An estimator $\hat{g} = \hat{g}(X)$ is said to be *unbiased* if

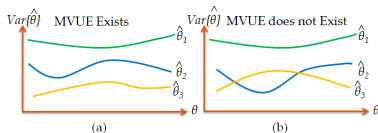
$$\mathbb{E}_\theta[\hat{g}(X)] = g(\theta), \quad \forall \theta \in \Theta$$

Uniformly minimum-variance unbiased estimator

- In statistics, a classic notion of an *optimal* frequentist estimator is *uniformly minimum-variance unbiased estimator (UMVUE)*
- An estimator $\hat{g} = \hat{g}(X)$ is said to be *unbiased* if

$$\mathbb{E}_\theta[\hat{g}(X)] = g(\theta), \quad \forall \theta \in \Theta$$

- Among all *unbiased* estimators, we naturally prefer those with smaller *variances*. An UMVUE, *if it exists*, is an *unbiased* estimator with *smaller or equal* variances than *any* other unbiased estimator of $g(\theta)$ for *all* possible values of θ :

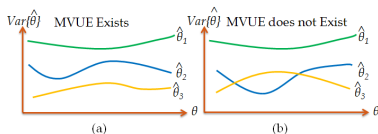


Uniformly minimum-variance unbiased estimator

- In statistics, a classic notion of an *optimal* frequentist estimator is *uniformly minimum-variance unbiased estimator (UMVUE)*
- An estimator $\hat{g} = \hat{g}(X)$ is said to be *unbiased* if

$$\mathbb{E}_\theta[\hat{g}(X)] = g(\theta), \quad \forall \theta \in \Theta$$

- Among all *unbiased* estimators, we naturally prefer those with smaller *variances*. An UMVUE, *if it exists*, is an *unbiased* estimator with *smaller or equal* variances than *any* other unbiased estimator of $g(\theta)$ for *all* possible values of θ :



- Note that in studying UMVUE, we may assume without out loss of generality that $g(\theta)$ is a *real-valued* function

Finding an UMVUE

- Unfortunately, for a general likelihood model $f_{\theta}(x)$, there is *no* systematic procedure for finding an UMVUE

Finding an UMVUE

- Unfortunately, for a general likelihood model $f_{\theta}(x)$, there is *no* systematic procedure for finding an UMVUE
- However, if the distribution family $f_{\theta}(x)$ has a *complete* sufficient statistic, an UMVUE may be found by applying the **Rao-Blackwell theorem** with the *complete* sufficient statistic

Rao-Blackwell theorem

- (*Rao-Blackwell theorem*): Let Y be a real-valued random variable and let Z be a random vector jointly distributed with Y . If we let

$$u(Z) := \mathbb{E}[Y|Z],$$

we have

$$\mathbb{E}[u(Z)] = \mathbb{E}[Y]$$

and

$$\text{Var}[u(Z)] \leq \text{Var}[Y]$$

Rao-Blackwell theorem

- (*Rao-Blackwell theorem*): Let Y be a real-valued random variable and let Z be a random vector jointly distributed with Y . If we let

$$u(Z) := \mathbb{E}[Y|Z],$$

we have

$$\mathbb{E}[u(Z)] = \mathbb{E}[Y]$$

and

$$\text{Var}[u(Z)] \leq \text{Var}[Y]$$

- (Proof) By the law of total expectation,

$$\mathbb{E}[u(Z)] = \mathbb{E}[\mathbb{E}[Y|Z]] = \mathbb{E}[Y]$$

Rao-Blackwell theorem

- (*Rao-Blackwell theorem*): Let Y be a real-valued random variable and let Z be a random vector jointly distributed with Y . If we let

$$u(Z) := \mathbb{E}[Y|Z],$$

we have

$$\mathbb{E}[u(Z)] = \mathbb{E}[Y]$$

and

$$\text{Var}[u(Z)] \leq \text{Var}[Y]$$

- (Proof) By the law of total expectation,

$$\mathbb{E}[u(Z)] = \mathbb{E}[\mathbb{E}[Y|Z]] = \mathbb{E}[Y]$$

- Next, let $\mathbb{E}[u(Z)] = \mathbb{E}[Y] = \mu$. We have

$$\text{Var}[Y]$$

$$= \mathbb{E}[(Y - \mu)^2] = \mathbb{E}[(Y - u(Z) + (u(Z) - \mu))^2]$$

$$= \mathbb{E}[(Y - u(Z))^2] + \mathbb{E}[(u(Z) - \mu)^2] + 2\mathbb{E}[(Y - u(Z))(u(Z) - \mu)]$$

- Note that $\mathbb{E}[u(Z)] = \mu$ so

$$\mathbb{E}[(u(Z) - \mu)^2] = \text{Var}[u(Z)]$$

- Note that $\mathbb{E}[u(Z)] = \mu$ so

$$\mathbb{E}[(u(Z) - \mu)^2] = \text{Var}[u(Z)]$$

- Furthermore, by the law of total expectation (again), we have

$$\begin{aligned}\mathbb{E}[(Y - u(Z))(u(Z) - \mu)] &= \mathbb{E}[\mathbb{E}[(Y - u(Z))(u(Z) - \mu)]|Z] \\ &= \mathbb{E}[(\mathbb{E}[Y|Z] - u(Z))(u(Z) - \mu)] \\ &= \mathbb{E}[(u(Z) - u(Z))(u(Z) - \mu)] \\ &= 0\end{aligned}$$

- Note that $\mathbb{E}[u(Z)] = \mu$ so

$$\mathbb{E}[(u(Z) - \mu)^2] = \text{Var}[u(Z)]$$

- Furthermore, by the law of total expectation (again), we have

$$\begin{aligned}\mathbb{E}[(Y - u(Z))(u(Z) - \mu)] &= \mathbb{E}[\mathbb{E}[(Y - u(Z))(u(Z) - \mu)]|Z] \\ &= \mathbb{E}[(\mathbb{E}[Y|Z] - u(Z))(u(Z) - \mu)] \\ &= \mathbb{E}[(u(Z) - u(Z))(u(Z) - \mu)] \\ &= 0\end{aligned}$$

- We thus have

$$\text{Var}[Y] = \mathbb{E}[(Y - g(Z))^2] + \text{Var}[g(Z)] \geq \text{Var}[g(Z)]$$

Rao-Blackwellization

- Consider $X \sim f_\theta(x)$. Let $\tilde{g} = \tilde{g}(X)$ be an *unbiased* estimator for $g(\theta)$, and let $T = T(X)$ be a *sufficient* statistic of θ

Rao-Blackwellization

- Consider $X \sim f_\theta(x)$. Let $\tilde{g} = \tilde{g}(X)$ be an *unbiased* estimator for $g(\theta)$, and let $T = T(X)$ be a *sufficient* statistic of θ
- Apply the *Rao-Blackwell theorem* with

$$Y = \tilde{g} \quad \text{and} \quad Z = T$$

Rao-Blackwellization

- Consider $X \sim f_\theta(x)$. Let $\tilde{g} = \tilde{g}(X)$ be an *unbiased* estimator for $g(\theta)$, and let $T = T(X)$ be a *sufficient* statistic of θ
- Apply the *Rao-Blackwell theorem* with

$$Y = \tilde{g} \quad \text{and} \quad Z = T$$

- Consider the new estimator

$$\hat{g} = \hat{g}(T) = \mathbb{E}_\theta[\tilde{g}|T]$$

where the function $\hat{g}$ does *not* depend on θ due to the *sufficiency* of T

Rao-Blackwellization

- Consider $X \sim f_\theta(x)$. Let $\tilde{g} = \tilde{g}(X)$ be an *unbiased* estimator for $g(\theta)$, and let $T = T(X)$ be a *sufficient* statistic of θ
- Apply the *Rao-Blackwell theorem* with

$$Y = \tilde{g} \quad \text{and} \quad Z = T$$

- Consider the new estimator

$$\hat{g} = \hat{g}(T) = \mathbb{E}_\theta[\tilde{g}|T]$$

where the function $\hat{g}$ does *not* depend on θ due to the *sufficiency* of T

- By the *Rao-Blackwell theorem*, we have:

- $\mathbb{E}_\theta[\hat{g}] = \mathbb{E}_\theta[\tilde{g}] = g(\theta)$
- $\text{Var}_\theta[\hat{g}] \leq \text{Var}_\theta[\tilde{g}]$

for all θ

- In words, if $\tilde{g}$ is an *unbiased* estimator of $g(\theta)$, then smoothing $\tilde{g}$ with respect to a *sufficient* statistic T decreases the variance while preserving the unbiasedness

- In words, if $\tilde{g}$ is an *unbiased* estimator of $g(\theta)$, then smoothing $\tilde{g}$ with respect to a *sufficient* statistic T decreases the variance while preserving the unbiasedness
- The Rao-Blackwell theorem tells us how to decrease the variance of an unbiased estimator. But when can we know that we get an *UMVUE*?

- In words, if $\tilde{g}$ is an *unbiased* estimator of $g(\theta)$, then smoothing $\tilde{g}$ with respect to a *sufficient* statistic T decreases the variance while preserving the unbiasedness
- The Rao-Blackwell theorem tells us how to decrease the variance of an unbiased estimator. But when can we know that we get an *UMVUE*?
- A short answer is when T is a *complete* sufficient statistic

Lehmann–Scheffé theorem

- (*Lehmann–Scheffé theorem*) If T is a *complete* statistic, then there is *at most one* unbiased estimator of $g(\theta)$ that is a function of T

Lehmann–Scheffé theorem

- (*Lehmann–Scheffé theorem*) If T is a *complete* statistic, then there is *at most one* unbiased estimator of $g(\theta)$ that is a function of T
- (Proof) Let $\hat{g}_1 = \hat{g}_1(T)$ and $\hat{g}_2 = \hat{g}_2(T)$, and assume that

$$\mathbb{E}_\theta[\hat{g}_1] = \mathbb{E}_\theta[\hat{g}_2] = g(\theta), \quad \forall \theta \in \Theta$$

Lehmann–Scheffé theorem

- (*Lehmann–Scheffé theorem*) If T is a *complete* statistic, then there is *at most one* unbiased estimator of $g(\theta)$ that is a function of T
- (Proof) Let $\hat{g}_1 = \hat{g}_1(T)$ and $\hat{g}_2 = \hat{g}_2(T)$, and assume that

$$\mathbb{E}_\theta[\hat{g}_1] = \mathbb{E}_\theta[\hat{g}_2] = g(\theta), \quad \forall \theta \in \Theta$$

- Let $\phi(t) := \hat{g}_1(t) - \hat{g}_2(t)$. We have

$$\mathbb{E}_\theta[\phi(T)] = \mathbb{E}_\theta[\hat{g}_1] - \mathbb{E}_\theta[\hat{g}_2] = g(\theta) - g(\theta) = 0, \quad \forall \theta \in \Theta$$

Lehmann–Scheffé theorem

- (*Lehmann–Scheffé theorem*) If T is a *complete* statistic, then there is *at most one* unbiased estimator of $g(\theta)$ that is a function of T
- (Proof) Let $\hat{g}_1 = \hat{g}_1(T)$ and $\hat{g}_2 = \hat{g}_2(T)$, and assume that

$$\mathbb{E}_\theta[\hat{g}_1] = \mathbb{E}_\theta[\hat{g}_2] = g(\theta), \quad \forall \theta \in \Theta$$

- Let $\phi(t) := \hat{g}_1(t) - \hat{g}_2(t)$. We have

$$\mathbb{E}_\theta[\phi(T)] = \mathbb{E}_\theta[\hat{g}_1] - \mathbb{E}_\theta[\hat{g}_2] = g(\theta) - g(\theta) = 0, \quad \forall \theta \in \Theta$$

- By the *completeness* of T , we have $\mathbb{P}_\theta[\phi(T) = 0] = 1$, i.e, $\hat{g}_1 = \hat{g}_2$ with probability 1

Finding UMVUE via complete sufficient statistics

- The *Lehmann–Scheffé* theorem suggests the following recipe for finding an UMVUE:

1) Find a *complete sufficient statistic* $T = T(X)$

2a) Find an unbiased estimator $\tilde{g} = \tilde{g}(X)$ of $g(\theta)$ and compute

$$\hat{g} = \hat{g}(T) = \mathbb{E}_{\theta}[\tilde{g}|T]$$

2b) Find a function $\hat{g}$ such that

$$\hat{g} = \hat{g}(T)$$

is an *unbiased* estimator of $g(\theta)$

Finding UMVUE via complete sufficient statistics

- The *Lehmann–Scheffé* theorem suggests the following recipe for finding an UMVUE:

1) Find a *complete sufficient statistic* $T = T(X)$

2a) Find an unbiased estimator $\tilde{g} = \tilde{g}(X)$ of $g(\theta)$ and compute

$$\hat{g} = \hat{g}(T) = \mathbb{E}_{\theta}[\tilde{g}|T]$$

2b) Find a function $\hat{g}$ such that

$$\hat{g} = \hat{g}(T)$$

is an *unbiased* estimator of $g(\theta)$

- (Theorem) If $\hat{g}$ is constructed by the recipe above, it is an UMVUE

Finding UMVUE via complete sufficient statistics

- The *Lehmann–Scheffé* theorem suggests the following recipe for finding an UMVUE:

1) Find a *complete sufficient statistic* $T = T(X)$

2a) Find an unbiased estimator $\tilde{g} = \tilde{g}(X)$ of $g(\theta)$ and compute

$$\hat{g} = \hat{g}(T) = \mathbb{E}_\theta[\tilde{g}|T]$$

2b) Find a function $\hat{g}$ such that

$$\hat{g} = \hat{g}(T)$$

is an *unbiased* estimator of $g(\theta)$

- (Theorem) If $\hat{g}$ is constructed by the recipe above, it is an UMVUE
- (Proof) Note that in either construction, $\hat{g}$ is a function of T such that $\hat{g} = \hat{g}(T)$ is an unbiased estimator of $g(\theta)$. Let $\tilde{g}' = \tilde{g}'(X)$ be *any* unbiased estimator of $g(\theta)$, it suffices to show that

$$\text{Var}_\theta[\hat{g}] \leq \text{Var}_\theta[\tilde{g}']$$

- Define

$$\hat{g}' = \hat{g}'(T) = \mathbb{E}_\theta[\tilde{g}'|T]$$

- Define

$$\hat{g}' = \hat{g}'(T) = \mathbb{E}_\theta[\tilde{g}'|T]$$

- By the *Rao-Blackwell* theorem, we have

$$\mathbb{E}_\theta[\hat{g}'] = \mathbb{E}_\theta[\tilde{g}'] = g(\theta) \quad \text{and} \quad \text{Var}_\theta[\hat{g}'] \leq \text{Var}_\theta[\tilde{g}']$$

for all $\theta \in \Theta$

- Define

$$\hat{g}' = \hat{g}'(T) = \mathbb{E}_\theta[\tilde{g}'|T]$$

- By the *Rao-Blackwell* theorem, we have

$$\mathbb{E}_\theta[\hat{g}'] = \mathbb{E}_\theta[\tilde{g}'] = g(\theta) \quad \text{and} \quad \text{Var}_\theta[\hat{g}'] \leq \text{Var}_\theta[\tilde{g}']$$

for all $\theta \in \Theta$

- Note that both $\hat{g}$ and $\hat{g}'$ are unbiased and functions of T . By the *Lehmann-Scheffé* theorem, $\hat{g} = \hat{g}'$ with probability 1

- Define

$$\hat{g}' = \hat{g}'(T) = \mathbb{E}_\theta[\tilde{g}'|T]$$

- By the *Rao-Blackwell* theorem, we have

$$\mathbb{E}_\theta[\hat{g}'] = \mathbb{E}_\theta[\tilde{g}'] = g(\theta) \quad \text{and} \quad \text{Var}_\theta[\hat{g}'] \leq \text{Var}_\theta[\tilde{g}']$$

for all $\theta \in \Theta$

- Note that both $\hat{g}$ and $\hat{g}'$ are unbiased and functions of T . By the *Lehmann-Scheffé* theorem, $\hat{g} = \hat{g}'$ with probability 1
- We thus have

$$\text{Var}_\theta[\hat{g}] = \text{Var}_\theta[\hat{g}'] \leq \text{Var}_\theta[\tilde{g}']$$

for all $\theta \in \Theta$, which completes the proof that $\hat{g}$ is an UMVUE for $g(\theta)$

Example: Bernoulli trials

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n X_i$$

Example: Bernoulli trials

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n X_i$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimator that is a function of T is an UMVUE for the unknown parameter θ

Example: Bernoulli trials

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n X_i$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimator that is a function of T is an UMVUE for the unknown parameter θ
- Note that

$$\mathbb{E}_{\theta}[T] = \sum_{i=1}^n \mathbb{E}_{\theta}[X_i] = n\theta,$$

so the *empirical mean*

$$\hat{\theta} = \frac{1}{n}T = \frac{1}{n} \sum_{i=1}^n X_i$$

is unbiased and hence is an UMVUE for θ

- The variance of $\hat{\theta}$ is given by:

$$\begin{aligned}\text{Var}_{\theta}[\hat{\theta}] &= \text{Var}_{\theta} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}_{\theta}[X_i] \\ &= \frac{1}{n^2} (n\theta(1 - \theta)) \\ &= \frac{\theta(1 - \theta)}{n},\end{aligned}$$

which decreases *linearly* with the sample size n

Example: Gaussian with unknown mean

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n X_i$$

Example: Gaussian with unknown mean

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n X_i$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimator that is a function of T is an UMVUE for the unknown parameter μ

Example: Gaussian with unknown mean

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n X_i$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimator that is a function of T is an UMVUE for the unknown parameter μ
- Note that

$$\mathbb{E}_{\mu}[T] = \sum_{i=1}^n \mathbb{E}_{\mu}[X_i] = n\mu,$$

so the *empirical mean*

$$\hat{\mu} = \frac{1}{n}T = \frac{1}{n} \sum_{i=1}^n X_i$$

is unbiased and hence is an UMVUE for μ

- The variance of $\hat{\mu}$ is given by:

$$\begin{aligned}\text{Var}_{\mu}[\hat{\theta}] &= \text{Var}_{\mu} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}_{\mu}[X_i] \\ &= \frac{1}{n^2} (n\sigma^2) \\ &= \frac{\sigma^2}{n},\end{aligned}$$

which decreases *linearly* with the sample size n

Example: Gaussian with unknown variance

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n (X_i - \mu)^2$$

Example: Gaussian with unknown variance

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n (X_i - \mu)^2$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimator that is a function of T is an UMVUE for the unknown parameter σ^2

Example: Gaussian with unknown variance

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic

$$T = \sum_{i=1}^n (X_i - \mu)^2$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimator that is a function of T is an UMVUE for the unknown parameter σ^2
- Note that

$$\mathbb{E}_{\sigma^2}[T] = \sum_{i=1}^n \mathbb{E}_{\sigma^2}[(X_i - \mu)^2] = n\sigma^2,$$

so the *empirical variance*

$$\widehat{\sigma^2} = \frac{1}{n}T = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

is unbiased and hence is an UMVUE for σ^2

- The variance of $\widehat{\sigma^2}$ is given by:

$$\begin{aligned}\text{Var}_{\sigma^2}[\hat{\theta}] &= \text{Var}_{\sigma^2} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 \right] \\&= \text{Var}_{\sigma^2} \left[\frac{\sigma^2}{n} \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \right] \\&= \frac{\sigma^4}{n^2} \text{Var}_{\sigma^2} \left[\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \right] \\&= \frac{\sigma^4}{n^2} (2n) \\&= \frac{2\sigma^4}{n},\end{aligned}$$

which decreases *linearly* with the sample size n

Example: Gaussian with unknown mean and variance

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic $T = (T_1, T_2)$, where

$$T_1 = \sum_{i=1}^n X_i \quad \text{and} \quad T_2 = \sum_{i=1}^n X_i^2$$

Example: Gaussian with unknown mean and variance

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic $T = (T_1, T_2)$, where

$$T_1 = \sum_{i=1}^n X_i \quad \text{and} \quad T_2 = \sum_{i=1}^n X_i^2$$

- Thus by the *Lehmann–Scheffé* theorem, any unbiased estimators that are a function of $T = (T_1, T_2)$ are UMVUE for the unknown parameters μ and σ^2

Example: Gaussian with unknown mean and variance

- Recall that in this case, the likelihood function is an *exponential* family of distributions with a *complete* sufficient statistic $T = (T_1, T_2)$, where

$$T_1 = \sum_{i=1}^n X_i \quad \text{and} \quad T_2 = \sum_{i=1}^n X_i^2$$

- Thus by the *Lehmann-Scheffé* theorem, any unbiased estimators that are a function of $T = (T_1, T_2)$ are UMVUE for the unknown parameters μ and σ^2
- To find an unbiased estimator of μ that is a function of T , simply notice that

$$\mathbb{E}_\theta[T_1] = \mathbb{E}_\theta \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n \mathbb{E}_\theta [X_i] = n\mu$$

- Therefore, the empirical mean

$$\hat{\mu} = \frac{1}{n}T_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

is an unbiased estimator of μ and hence is an UMVUE of μ

- Therefore, the empirical mean

$$\hat{\mu} = \frac{1}{n}T_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

is an unbiased estimator of μ and hence is an UMVUE of μ

- Recall that the same empirical-mean estimator is also an UMVUE of μ when the variance σ^2 is *known*, and the variance of the estimator is given by

$$\text{Var}_{\theta}[\hat{\mu}] = \frac{\sigma^2}{n},$$

which decreases *linearly* with the sample size n

- Therefore, the empirical mean

$$\hat{\mu} = \frac{1}{n}T_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

is an unbiased estimator of μ and hence is an UMVUE of μ

- Recall that the same empirical-mean estimator is also an UMVUE of μ when the variance σ^2 is *known*, and the variance of the estimator is given by

$$\text{Var}_{\theta}[\hat{\mu}] = \frac{\sigma^2}{n},$$

which decreases *linearly* with the sample size n

- To find an UMVUE for σ^2 , recall that when the mean parameter μ is the *known*, an UMVUE is given by the *empirical variance*

$$\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

- If μ is *unknown*, it is very tempting to replace μ by its UMVUE $\hat{\mu}$ and consider the estimator

$$\widehat{\sigma^2}' = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2,$$

where

$$\hat{\mu} = \frac{1}{n} \sum_{j=1}^n X_j$$

- If μ is *unknown*, it is very tempting to replace μ by its UMVUE $\hat{\mu}$ and consider the estimator

$$\widehat{\sigma^2}' = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2,$$

where

$$\hat{\mu} = \frac{1}{n} \sum_{j=1}^n X_j$$

- Let us first check if the above estimator is *unbiased* for σ^2 .

- Let $X_i = \sigma Z_i + \mu$, where $(Z_1, \dots, Z_n)$ are i.i.d. standard Gaussian variables. We can write

$$\begin{aligned}\widehat{\sigma^2}' &= \frac{1}{n} \sum_{i=1}^n \left(X_i - \frac{1}{n} \sum_{j=1}^n X_j \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n \left((\sigma Z_i + \mu) - \frac{1}{n} \sum_{j=1}^n (\sigma Z_j + \mu) \right)^2 \\ &= \frac{\sigma^2}{n} \sum_{i=1}^n \left(Z_i - \frac{1}{n} \sum_{j=1}^n Z_j \right)^2\end{aligned}$$

- Let $X_i = \sigma Z_i + \mu$, where $(Z_1, \dots, Z_n)$ are i.i.d. standard Gaussian variables. We can write

$$\begin{aligned}\widehat{\sigma^2}' &= \frac{1}{n} \sum_{i=1}^n \left(X_i - \frac{1}{n} \sum_{j=1}^n X_j \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n \left((\sigma Z_i + \mu) - \frac{1}{n} \sum_{j=1}^n (\sigma Z_j + \mu) \right)^2 \\ &= \frac{\sigma^2}{n} \sum_{i=1}^n \left(Z_i - \frac{1}{n} \sum_{j=1}^n Z_j \right)^2\end{aligned}$$

- (*Cochran's theorem*)

$$\sum_{i=1}^n \left(Z_i - \frac{1}{n} \sum_{j=1}^n Z_j \right)^2$$

is a *Chi-squared* random variable with degree of freedom $n - 1$

- We thus have

$$\mathbb{E}_\theta \left[\widehat{\sigma^2}' \right] = \frac{\sigma^2}{n} (n-1) = \frac{n-1}{n} \sigma^2$$

- We thus have

$$\mathbb{E}_\theta \left[\widehat{\sigma^2}' \right] = \frac{\sigma^2}{n}(n-1) = \frac{n-1}{n}\sigma^2$$

- Note that while the above estimator is *biased* for σ^2 , we can easily get rid of the bias by multiplying $\widehat{\sigma^2}'$ with the simple scaling factor $\frac{n}{n-1}$ and get an *unbiased* estimator for σ^2 :

$$\widehat{\sigma^2}'' = \frac{n}{n-1}\widehat{\sigma^2}' = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

- We thus have

$$\mathbb{E}_\theta \left[\widehat{\sigma^2}' \right] = \frac{\sigma^2}{n} (n-1) = \frac{n-1}{n} \sigma^2$$

- Note that while the above estimator is *biased* for σ^2 , we can easily get rid of the bias by multiplying $\widehat{\sigma^2}'$ with the simple scaling factor $\frac{n}{n-1}$ and get an *unbiased* estimator for σ^2 :

$$\widehat{\sigma^2}'' = \frac{n}{n-1} \widehat{\sigma^2}' = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

- Is $\widehat{\sigma^2}''$ a function of T ?

- Note that

$$\begin{aligned}\sum_{i=1}^n (X_i - \hat{\mu})^2 &= \sum_{i=1}^n X_i^2 - 2\hat{\mu} \sum_{i=1}^n X_i + n\hat{\mu}^2 \\ &= T_2 - 2\left(\frac{T_1}{n}\right) T_1 + n\left(\frac{T_1}{n}\right)^2 \\ &= T_2 - \frac{T_1^2}{n} \\ &= \frac{nT_2 - T_1^2}{n}\end{aligned}$$

which is a function of $T = (T_1, T_2)$ and hence an UMVUE of σ^2

- Note that

$$\begin{aligned}\sum_{i=1}^n (X_i - \hat{\mu})^2 &= \sum_{i=1}^n X_i^2 - 2\hat{\mu} \sum_{i=1}^n X_i + n\hat{\mu}^2 \\ &= T_2 - 2 \left(\frac{T_1}{n} \right) T_1 + n \left(\frac{T_1}{n} \right)^2 \\ &= T_2 - \frac{T_1^2}{n} \\ &= \frac{nT_2 - T_1^2}{n}\end{aligned}$$

- We thus have

$$\widehat{\sigma^2}'' = \frac{nT_2 - T_1^2}{n(n-1)},$$

which is a function of $T = (T_1, T_2)$ and hence an UMVUE of σ^2

- The variance of $\widehat{\sigma^2}''$ can be calculated as:

$$\begin{aligned}\text{Var}_\theta \left[\widehat{\sigma^2}'' \right] &= \text{Var}_\theta \left[\frac{\sigma^2}{n-1} \sum_{i=1}^n \left(Z_i - \frac{1}{n} \sum_{j=1}^n Z_j \right)^2 \right] \\ &= \frac{\sigma^4}{(n-1)^2} \text{Var}_\theta \left[\sum_{i=1}^n \left(Z_i - \frac{1}{n} \sum_{j=1}^n Z_j \right)^2 \right] \\ &= \frac{\sigma^4}{(n-1)^2} (2(n-1)) \\ &= \frac{2\sigma^4}{n-1}\end{aligned}$$

which is *strictly* greater than

$$\text{Var}_\theta \left[\widehat{\sigma^2} \right] = \frac{2\sigma^4}{n}$$

- This is the price to pay for *not* knowing the value of the mean parameter μ

- This is the price to pay for *not* knowing the value of the mean parameter μ
- Finally, note that the UMVUE

$$\widehat{\sigma^2}'' = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

does *not* work for $n = 1$. What happens to $n = 1$?

- This is the price to pay for *not* knowing the value of the mean parameter μ
- Finally, note that the UMVUE

$$\widehat{\sigma^2}'' = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

does *not* work for $n = 1$. What happens to $n = 1$?

- When $n = 1$, X itself is a complete sufficient statistic. Therefore, if an *unbiased* estimator exists, it is also an UMVUE. As an exercise, show that for $n = 1$, an unbiased estimator for σ^2 does *not* exist (when both μ and σ^2 are unknown)

- This is the price to pay for *not* knowing the value of the mean parameter μ
- Finally, note that the UMVUE

$$\widehat{\sigma^2}'' = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

does *not* work for $n = 1$. What happens to $n = 1$?

- When $n = 1$, X itself is a complete sufficient statistic. Therefore, if an *unbiased* estimator exists, it is also an UMVUE. As an exercise, show that for $n = 1$, an unbiased estimator for σ^2 does *not* exist (when both μ and σ^2 are unknown)
- We can also conclude from the above example that even when a complete sufficient exists, an UMVUE may *not* exist

Example: Uniform with unknown support

- Recall that in this case,

$$T = \max_{i \in [n]} X_i$$

is a *complete* sufficient statistic for θ

Example: Uniform with unknown support

- Recall that in this case,

$$T = \max_{i \in [n]} X_i$$

is a *complete* sufficient statistic for θ

- To find an unbiased estimator that is a function of T , let us first calculate the *expectation* of T

Example: Uniform with unknown support

- Recall that in this case,

$$T = \max_{i \in [n]} X_i$$

is a *complete* sufficient statistic for θ

- To find an unbiased estimator that is a function of T , let us first calculate the *expectation* of T
- Recall that the pdf of T can be calculated as

$$f_{\theta}(t) = nt^{n-1}\theta^{-n}1_{\{0 \leq t \leq \theta\}}$$

Example: Uniform with unknown support

- Recall that in this case,

$$T = \max_{i \in [n]} X_i$$

is a *complete* sufficient statistic for θ

- To find an unbiased estimator that is a function of T , let us first calculate the *expectation* of T
- Recall that the pdf of T can be calculated as

$$f_{\theta}(t) = nt^{n-1}\theta^{-n}1_{\{0 \leq t \leq \theta\}}$$

- The expectation of T is thus given by:

$$\mathbb{E}_{\theta}[T] = \int_0^{\theta} tf_{\theta}(t)dt = n \int_0^{\theta} \left(\frac{t}{\theta}\right)^n dt = n\theta \int_0^1 \beta^n d\beta = \frac{n\theta}{n+1}$$

- Thus, if we let

$$\hat{\theta} = \frac{n+1}{n}T,$$

it is *unbiased* and hence must be an *UMVUE* for θ

- Thus, if we let

$$\hat{\theta} = \frac{n+1}{n}T,$$

it is *unbiased* and hence must be an *UMVUE* for θ

- The variance of $\hat{\theta}$ can be calculated as

$$\text{Var}_{\theta}[\hat{\theta}] = \text{Var}_{\theta}\left[\frac{n+1}{n}T\right] = \left(\frac{n+1}{n}\right)^2 \text{Var}_{\theta}[T]$$

- Thus, if we let

$$\hat{\theta} = \frac{n+1}{n}T,$$

it is *unbiased* and hence must be an *UMVUE* for θ

- The variance of $\hat{\theta}$ can be calculated as

$$\text{Var}_{\theta}[\hat{\theta}] = \text{Var}_{\theta}\left[\frac{n+1}{n}T\right] = \left(\frac{n+1}{n}\right)^2 \text{Var}_{\theta}[T]$$

- Note that

$$\begin{aligned}\mathbb{E}_{\theta}[T^2] &= \int_0^{\theta} t^2 f_{\theta}(t) dt \\ &= n\theta \int_0^{\theta} \left(\frac{t}{\theta}\right)^{n+1} dt \\ &= n\theta^2 \int_0^1 \beta^{n+1} d\beta = \frac{n\theta^2}{n+2}\end{aligned}$$

- It follows that

$$\begin{aligned}\mathrm{Var}_\theta[T] &= \mathbb{E}_\theta[T^2] - (\mathbb{E}_\theta[T])^2 \\ &= \frac{n\theta^2}{n+2} - \left(\frac{n\theta}{n+1}\right)^2 \\ &= \frac{n(n+1)^2 - n^2(n+2)}{(n+1)^2(n+2)}\theta^2 \\ &= \frac{n\theta^2}{(n+1)^2(n+2)}\end{aligned}$$

and hence

$$\mathrm{Var}_\theta[\hat{\theta}] = \left(\frac{n+1}{n}\right)^2 \mathrm{Var}_\theta[T] = \frac{\theta^2}{n(n+2)},$$

which decreases *quadratically* with the sample size n